Singular value decomposition (SVD) is a highlight of linear algebra. We can find almost all basic (but important) concepts in it.

unlike diagonalizing decomposition, SVD works for all matrix $A \in \mathbb{R}^{n\times m}$

1. require square

2. not all matrix is diagnosible

What is SVD?

$$A = U \Sigma V^T$$

U: orthonormal, eigenvector of AA^T

Σ: diagonal matrix $\begin{pmatrix} \sigma_1 & & & & & & \\ & \sigma_2 & & & & & \\ & & \ddots & & & & \\ & & & \sigma_r & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{pmatrix}_{n\times m}$ → singular value. $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$

V^T → orthonormal → eigenvector of A^TA

How it is derived?

Note that AA^T is a semi-positive definite matrix, we must have

AA^T is diagonalzable, So $AA^T = U \Lambda U^T$, where $\Lambda = \begin{pmatrix} \sigma_{u_1} & & \\ & \ddots & \\ & & \sigma_{u_n} \end{pmatrix}$, $\sigma_{u_i} \geq 0$

U → orthonormal

Consider any eigenvector u_i of AA^T of non-zero eigen-value σ_{u_i}:

$$AA^T u_i = \sigma_{u_i} u_i$$

$$\Rightarrow u_i^T AA^T u_i = \sigma_{u_i} u_i^T u_i$$

$$\Rightarrow \|A^T u_i\|^2 = \sigma_{u_i}$$

also $A^TAA^T u_i = \sigma_{u_i} A^T u_i$

$\Rightarrow A^TA(A^T u_i) = \sigma_{u_i}(A^T u_i)$

$\Rightarrow$ $A^T u_i$ is a eigenvector of A^TA with corresponding eigenvalue σ_{u_i}

Denote $v_i = \dfrac{A^T u_i}{\|A^T u_i\|} = \dfrac{A^T u_i}{\sqrt{\sigma_{u_i}}}$

$\Rightarrow \underbrace{A^T u_i = v_i \sqrt{\sigma_{u_i}}}_{(*)} \Rightarrow AA^T u_i = A v_i \sqrt{\sigma_{u_i}} \Rightarrow \sqrt{\sigma_{u_i}}\, u_i = A v_i$

Hence, for any eigenvector u_i of A^TA with positive eigenvalues σ_{u_i}, there is a corresponding eigenvector v_i with the same eigenvalues such that

$$Av_i = \sqrt{\sigma_{u_i}}\, u_i$$

Similarly, we can show that for any v_i with positive eigenvalue has the corresponding u_i with the same eigenvalues.

$\sigma_1 \geq \cdots \geq \sigma_r > 0$

So we have $A(v_1, \ldots, v_r) = (u_1, \ldots, u_r) \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{pmatrix}$

here $\sigma_i = \sqrt{\sigma_{u_i}}$ is the square root of the eigenvalue of A^TA / (AA^T)

Why is r = rank of A?

1. $\text{rank}(A^TA) = \text{rank}(A)$

Since they have the same null space

if $Ax = 0 \Rightarrow A^TAx = 0$

if $A^TAx = 0 \Rightarrow x^TA^TAx = 0 \Rightarrow \|Ax\| = 0 \Rightarrow Ax = 0$

So they have the same row space $\Rightarrow \text{rank}(A^TA) = \text{rank}(A)$

$\Rightarrow \text{rank}(A) = \text{rank}(A^T)$
$= \text{rank}(AA^T)$

2. A^TA can be diagnolized:

rank = number of non-zero eigenvalues
= number of positive eigenvalues

Since $A^TA = U\Lambda U^T$

invertible matrix does not change rank

$\xrightarrow{\text{if } M^{-1} \text{ exist}}$ $Ax = 0 \Leftrightarrow MAx = 0$

if M is invertible A and MA share the same null space

$A(v_1, \dots, v_r) = \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{pmatrix} (u_1, \dots, u_r)$ Considers only eigenvectors with corresponding positive eigenvalues

Note that each $u_i \in Col(A)$

Also since (*), we have $A^T(u_1, \dots, u_r) = \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{pmatrix} (v_1, \dots, v_r)$

We have each $v_i \in col(A^T)$

So let $u_{r+1}, \dots, u_n$ be a orthonomal basis of $null(A^T)$

$v_{r+1}, \dots, v_m$ be a orthonomal basis of $null(A)$

Hence we further have

$$\underbrace{A}_{\in \mathbb{R}^{n\times m}} \overbrace{(v_1, \dots, v_r\ v_{r+1}, \dots, v_m)}^{\mathbb{R}^{m\times m}} = \overbrace{(u_1, \dots, u_r, u_{r+1}, \dots, u_n)}^{\mathbb{R}^{n\times n}} \underbrace{\left(\begin{array}{ccc|c} \overbrace{\begin{matrix}\sigma_1 & & \\ & \ddots & \\ & & \sigma_r\end{matrix}}^{r\times r} & & & \\ \hline & & & \underbrace{\begin{matrix}0 & & \\ & \ddots & \\ & & 0\end{matrix}}_{\text{padding } 0} \end{array}\right)}_{\mathbb{R}^{n\times m}}$$